```
from keras.layers import Input
from keras.layers import Dense
visible = Input(shape=(2,))
hidden = Dense(2)(visible)
visible = Input(shape=(2,))
hidden = Dense(2)(visible)
model = Model(inputs=visible, outputs=hidden)
visible = Input(shape=(10,))
hidden1 = Dense(10, activation='relu')(visible)
hidden2 = Dense(20, activation='relu')(hidden1)
hidden3 = Dense(10, activation='relu')(hidden2)
output = Dense(1, activation='sigmoid')(hidden3)
model = Model(inputs=visible, outputs=output)
# summarize layers
print(model.summary())
# plot graph
plot_model(model, to_file='multilayer_perceptron_graph.png')
model.compile(Adam(lr=0.05), 'binary_crossentropy', metrics=['accuracy'])

model_eval = model.evaluate(X_test, y_test)
plot_model(model, to_file="simple_keras_model.png", show_layer_names=True, show_shapes=True)
metadata_1 = y_classifier + np.random.gumbel(scale = 0.6, size = n_row)
metadata_2 = y_classifier - np.random.laplace(scale = 0.5, size = n_row)
metadata = np.array([metadata_1,metadata_2]).T

# Create training and test set
metadata_train = metadata[idx_train,:]
metadata_test = metadata[idx_test,:]
input_dat = Input(shape=(3,)) # for the three columns of dat_train
n_net_layer = Dense(50, activation='relu') # first dense layer
x1 = n_net_layer(input_dat)
x1 = Dropout(0.5)(x1)

input_metadata = Input(shape=(2,))
```